DO NOW — RECALL

Data capture, integrity and ACID transactions

Name: _____ Class: _____ Date: _____

1. How does DELETE differ from DROP TABLE?

2. What is a foreign key?

3. What does database consistency mean?

4. Array scores=[6,2,9,4]; replace element at index2 with element0+element1. Give final array and sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

DELETE removes row data; DROP removes table definition.

2. Model answer

Reference from one table to another key.

3. Model answer

Constraints remain satisfied by a valid committed transaction.

4. Model answer

scores[2]=6+2=8; array [6,2,8,4]; sum20.

Vocabulary — know these terms

Atomicity — All changes or none.

Consistency — Constraints preserved.

Isolation — Concurrent effects safely separated.

Durability — Committed changes persist.

referential integrity — Foreign keys refer to valid rows.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Check the answer

Explain which rule makes this result correct.

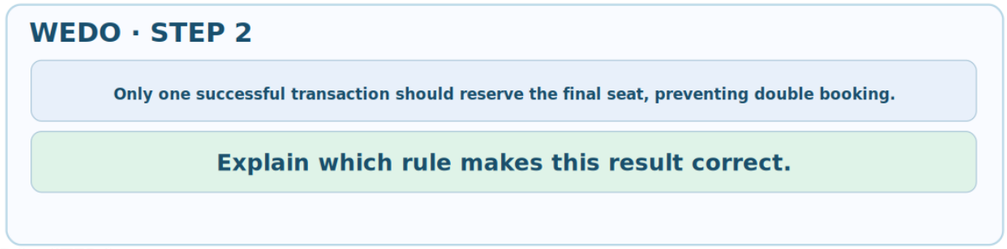
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two users attempt to reserve the last seat. Explain a safe database strategy.

Only one successful transaction should reserve the final seat, preventing double booking.



WEDO · STEP 2

NEW LEARNING

Read, annotate and explain

Understand

Data capture gathers data, while validation, verification and referential integrity address different correctness concerns.

Trace

Atomicity ensures a transaction is all-or-nothing; consistency preserves constraints; isolation addresses concurrent interference; durability keeps committed results.

Explain

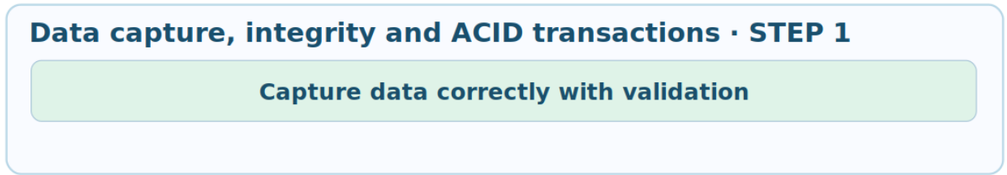
Row locking, transactions and appropriate constraints help prevent two concurrent changes from corrupting financial or stock records.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

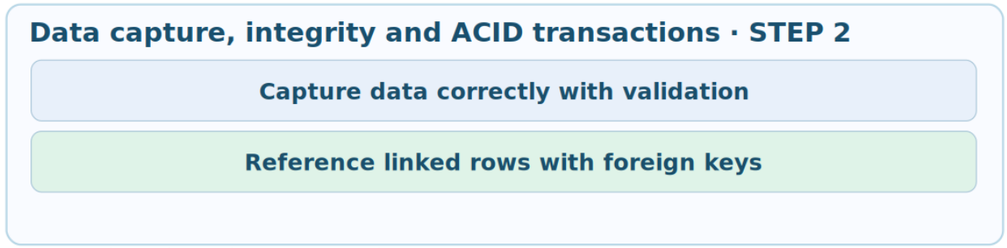
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



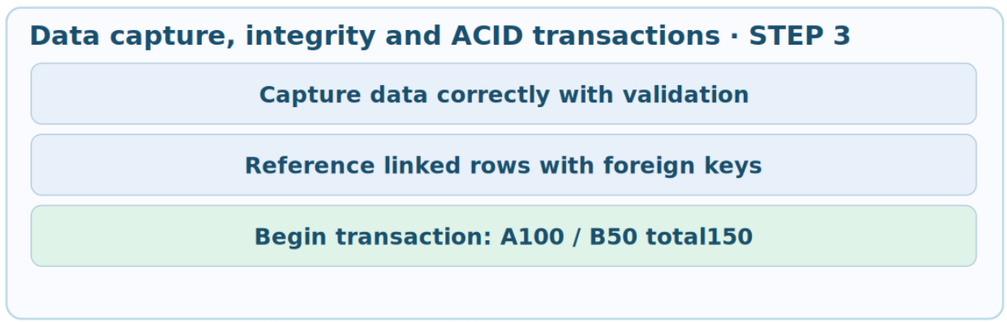
Data capture, integrity and ACID transactions · STEP 1

New Learning



Data capture, integrity and ACID transactions · STEP 2

New Learning



Data capture, integrity and ACID transactions · STEP 3

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A failed two-step bank transfer may safely leave the debit saved without the credit.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

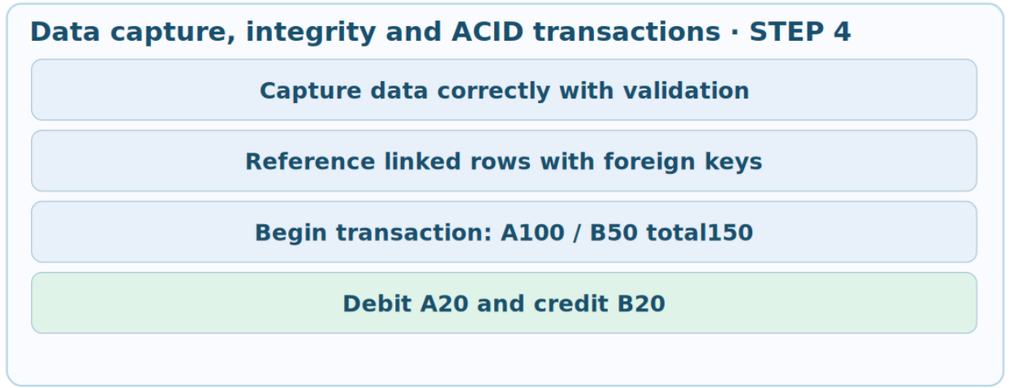
Question 3

Two clients try to book one remaining place. Explain how transaction isolation and locking protect integrity. [6]

New Learning · original lesson diagrams

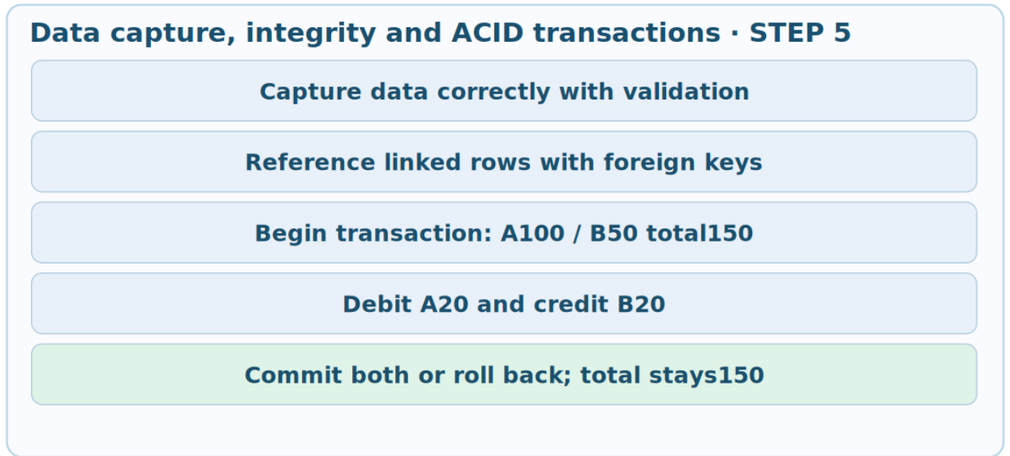
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Data capture, integrity and ACID transactions · STEP 4

New Learning



Data capture, integrity and ACID transactions · STEP 5

I DO — WORKED EXAMPLE

Follow the worked method

A stock purchase needs to reduce Stock and create Order together. Why use a transaction?

Predict

Both writes must succeed together or be undone so stock does not change without a valid order.

Check the answer

Explain which rule makes this result correct.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State the four ACID properties. [4]

Question 2

A transfer starts A100, B50. Move £20 and explain rollback on failure. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict Only one successful transaction should reserve the final seat, preventing double booking.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A stock purchase needs to reduce Stock and create Order together. Why use a transaction?

Both writes must succeed together or be undone so stock does not change without a valid order.



IDO · STEP 1

A stock purchase needs to reduce Stock and create Order together. Why use a transaction?

Both writes must succeed together or be undone so stock does not change without a valid order.



IDO · STEP 2

WE DO — GUIDED PRACTICE

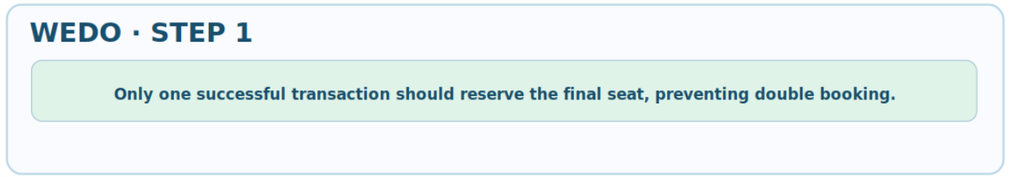
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Two users attempt to reserve the last seat. Explain a safe database strategy.

Only one successful transaction should reserve the final seat, preventing double booking.



WEDO · STEP 1